

RAGHAVENDRA PANDURANG JADHAV

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OBJECTIVE

Mechanical Engineering graduate with professional experience in large-scale cryogenic LPG infrastructure projects at BPCL and leadership experience in competitive robotics. Seeking an M.S. in Mechanical Engineering with a specialisation in Robotics to deepen my expertise in intelligent robotic systems, controls, and autonomous technologies, with the long-term goal of developing next-generation surgical robotic systems.

EDUCATION

National Institute of Technology Karnataka, Surathkal

Bachelor of Technology in Mechanical Engineering

With a Minor Degree in Electronics & Communication Engineering

Dec 2021 – May 2025

Major CGPA: 7.65/10

Minor CGPA: 7.40/10

ACADEMIC PROJECTS

Basketball Playing Robots [NITK Surathkal]

Aug 2024 – May 2025

Team Size: 40

Role: Team Lead

Description: Developed two human-scale robots capable of playing a 2v2 basketball match for the ABU Robocon 2025 competition. Delivered fully functional robots by designing a four-wheel holonomic drive with IMU- and encoder-based odometry for autonomous navigation, optimising a flywheel-based shooting mechanism, and developing a custom PCB architecture that reduced wiring complexity and improved system reliability. The project's development process involved mechanical design, PCB design, control system tuning, and manufacturing using 3D printing, laser cutting, CNC machining, manual machining, and soldering.

Autonomous Mobile Robots for Object Manipulation [NITK Surathkal]

Aug 2023 – July 2024

Team Size: 32

Role: Team Lead

Description: Designed and implemented autonomous four-wheel holonomic robots for the ABU Robocon 2024 competition, capable of navigating waypoints, picking up conduits, kicking footballs, and sorting objects autonomously. Successfully demonstrated fully functional robots by integrating encoder-based navigation, computer vision for colour-based object detection, and custom control algorithms. The project combined mechanical design (SolidWorks), custom PCB development (Altium Designer), and manufacturing techniques, including CNC machining, laser cutting, additive manufacturing, and PCB fabrication, to realise an end-to-end autonomous robotic system.

15KG Combat Robot – GNARLY [NITK Surathkal]

July– Oct 2023

Team Size: 15

Role: Team Lead

Description: Established NITK's first combat robotics program to design and develop a high-performance 15 kg combat robot for Engineer'23, the institute's flagship technical festival. Secured over \$3,000 in funding through a management proposal and developed one of India's few four-wheel-drive (4WD) combat robots in the 15 kg category, featuring a vertical bar spinner, chain-driven drivetrain, and belt-driven weapon system optimised for speed, impact, and durability. The project involved mechanical design in SolidWorks, drivetrain design, power transmission, structural optimisation, material selection (aluminium, titanium, and stainless steel), and precision manufacturing using CNC machining, laser cutting, turning, welding, and manual machining to develop a competition-ready combat robot.

Ring Shooting Holonomic Mobile Robots [NITK Surathkal]

July 2022 – June 2023

Team Size: 12

Role: Team Member

Description: Designed and developed two three-wheel holonomic robots for the ABU Robocon 2023 competition, capable of omnidirectional navigation, ring collection, and precision shooting at multiple target heights. Engineered a spring-loaded catapult, a dual-flywheel launcher with a custom gearbox, and two custom PCBs for power distribution and hardware integration, while implementing a NodeMCU-based wireless control system with custom navigation algorithms. Leveraged Fusion 360, PCB design, CNC milling, laser cutting, polymer 3D printing, TIG welding, soldering, and mechanical assembly to deliver a competition-ready robotic platform.

Pharma Bot [NITK Surathkal]

July 2022 – June 2023

Team Size: 12

Role: Team Lead

Description: Engineered an autonomous medicine-delivery robot by integrating OpenCV-based perception, QR code and ArUco marker localization, and autonomous navigation in a structured environment. Developed a vision-based localization framework using an overhead camera to estimate the robot's pose and synchronize its real-time state with CoppeliaSim for digital twin visualization, while implementing constrained path planning and task scheduling algorithms. Ranked among the Top 16 teams nationally in the e-Yantra Robotics Competition.

WORK EXPERIENCE

Bharat Petroleum Corporation Limited, Navi Mumbai

June 2025 – Present

Assistant Executive – Site Engineer

- Contribute to the successful execution of BPCL's refrigerated LPG import terminal expansion project worth USD 18.39 Million, increasing terminal handling capacity from 1.2 MMTPA to 3.0 MMTPA by coordinating mechanical installation, equipment integration, and construction activities while ensuring compliance with OISD standards, engineering specifications, and project schedules.
- Manage the design, statutory approvals, and multidisciplinary coordination for a USD 3.11 Million package comprising four railway overbridges, the critical path and primary bottleneck for the terminal expansion project, by collaborating with consultants, proof consultants, Central Railways, and internal BPCL stakeholders to resolve engineering and execution challenges.
- Oversee fabrication, quality control, and site execution of structural steel components, ensuring compliance with approved drawings, technical specifications, and construction standards.
- Execute equipment installation, hydrostatic and pneumatic testing, instrumentation hot-loop testing, and pre-commissioning activities for cryogenic LPG facilities, including storage tanks, pipelines, and associated mechanical systems.

INTERNSHIP

Summer Intern, Manufacturing Division, Bajaj Auto Limited, Pune

May – July 2024

- Analyzed assembly line workflow, work content distribution, and task sequencing across four motorcycle models to identify bottlenecks and opportunities for production line optimization.
- Performed time studies and line balancing by redistributing 9.25 minutes of work content across chassis overhead, volumetric, vehicle overhead, and slat conveyor systems to improve resource utilization.
- Increased production throughput from 360 to 546 vehicles per shift (51% improvement) by optimizing task allocation, eliminating bottlenecks, and balancing workloads across conveyor stations.
- Applied industrial engineering principles, production planning, work measurement, and process optimization techniques to improve assembly line efficiency and manufacturing productivity.

TECHNICAL SKILLS

- **Software and Programming:** SolidWorks, Fusion 360, AutoCAD, Ansys, Altium, Eagle, Ultimaker Cura, MATLAB and Simulink, Python, C/C++, Bash, Cloudflared
- **Hardware:** Raspberry Pi, Arduino, Soldering, PCB Manufacturing, 3D Printing, CNC Machining, Laser Cutting, Manual Machining

CERTIFICATIONS/WORKSHOPS/ADDITIONAL COURSES

Basics of Embedded Systems and Robotics | e-Yantra, IIT Bombay

July 2023

CSWA – Certified Solidworks Associate

Sept 2022

MATLAB Onramp | Mathworks

July 2022

SIMULINK Onramp | Mathworks

July 2022

Introduction to C++ | Coding Ninjas

July 2022

PATENTS

1. Dr. Khyati Verma, Dr. Siddharth R, Dr. Ajay Rai E, **Raghvendra Pandurang Jadhav**, Dhiren V Bhandary, Dr. Ganagadharan K V, Dr. Rohan Mascarenhas, "An automated ergonogauge device for measuring three-dimensional spatial data", Application No. 202541029107 (Filed on 27/03/2025) – *Currently under Examination*.
2. Dr. Priyalakshmi, Dr. Khyati Verma, Dhiren V Bhandary, **Raghvendra Pandurang Jadhav**, Dr. Ganagadharan K V, Dr. Praveena Shetty, Dr Greeshma Kannan, "Orthodontic Appliances to Prevent Tongue Thrusting Habit", Application No. 202541032533 (Filed on 02/04/2025) – *Currently under Examination*.

ACHIEVEMENTS/CO-CURRICULAR/EXTRACURRICULAR

- Secured second runner-up in Robowars at VIT Vellore's Gravitas 2024, serving as the primary robot driver during combat matches.
- Achieved runners-up in the IEEE NITK-hosted Embaddethon 2024, which dealt with an assignment focused on Computer Vision and Sensor Fusion.
- Qualified for the national round of ABU Robocon in 2023, 2024 & 2025, ranking among the top teams with scores of 96, 95 & 88 out of 100 in the Robot Design Documentation category.
- Advanced to the semi-finals of the IIT Bombay International Robowars competition in the under-15 kg category held during TECHFEST 2023.
- Earned runners-up in Robowars contest held during Manipal's Technical fest TECHTATVA'23.
- Won the Open Innovation Track at Technica'22, a national hackathon hosted by ISTE-VIT, competing against multidisciplinary student teams from across India.

LEADERSHIP ROLES

Vice-Captain – CSD Robocon NITK, NITK Surathkal July 2024 – May 2025

- Led a 40-member robotics team representing the institute at the international ABU Robocon 2025 competition.
- Managed team operations and finances, including fundraising, treasury management, procurement, and scheduling of activities.

Co-Founder & Lead – Team GNARLY, NITK Surathkal Sept 2023 – May 2025

- Led a 15-member combat robotics team, driving design, funding and participation in inter-university competitions.

Media & Web Head - CSD Robocon NITK, NITK Surathkal Jan 2023 – July 2024

- Revamped the team's digital presence by redesigning websites and outreach platforms.

COMMUNITY INVOLVEMENT

- Organized a robotics outreach showcase at ENGINEER'24 (NITK), engaging students and the public through interactive demonstrations of line-following, pick-and-place, and autonomous robotic systems to promote STEM education and inspire interest in robotics.
- Conducted a Knowledge Sharing Session – Aero Trivia, for 10th-class students about the Engineering behind Drones and their application, Aug 2021